Final-Term (Summer 2023-2024)

Dennis Zill - A first course in complex analysis with applications-Jones and Bartlett (2003)

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2	Chapter 6: Series and Residues	6.3: Laurent's Series		Ex 6.3	
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Complex Integrals

In Problems 1-16, evaluate the given integral along the indicated contour.

1. $\int_C (z+3)\,dz$, where C is $x = 2t$, $y = 4t - 1$, $1 \le t \le 3$

2. $\int_C (2\bar{z} - z)\,dz$, where C is $x = -t$, $y = t^2 + 2$, $0 \le t \le 2$

3. $\int_C z^2\,dz$, where C is $z(t) = 3t + 2it$, $-2 \le t \le 2$

4. $\int_C (3z^2 - 2z)\,dz$, where C is $z(t) = t + it^2$, $0 \le t \le 1$

5. $\int_C \frac{z+1}{z}\,dz$, where C is the right half of the circle $|z| = 1$ from $z = -i$ to $z = i$

6. $\int_C |z|^2\,dz$, where C is $x = t^2$, $y = 1/t$, $1 \le t < 2$

7. $\oint_C \operatorname{Re}(z)\,dz$, where C is the circle $|z| = 1$

8. $\oint_C \left(\frac{1}{(z+i)^3} - \frac{5}{z+i} + 8\right) dz$, where C is the circle $|z+i| = 1$, $0 \le t \le 2\pi$

9. $\int_C (x^2 + iy^3)\,dz$, where C is the straight line from $z = 1$ to $z = i$

10. $\int_C (x^2 - iy^3)\,dz$, where C is the lower half of the circle $|z| = 1$ from $z = -1$ to $z = 1$

11. $\int_C e^z\,dz$, where C is the polygonal path consisting of the line segments from $z = 0$ to $z = 2$ and from $z = 2$ to $z = 1 + \pi i$

12. $\int_C \sin z\,dz$, where C is the polygonal path consisting of the line segments from $z = 0$ to $z = 1$ and from $z = 1$ to $z = 1 + i$

13. $\int_C \operatorname{Im}(z - i)\,dz$, where C is the polygonal path consisting of the circular arc along $|z| = 1$ from $z = 1$ to $z = i$ and the line segment from $z = i$ to $z = -1$

14. $\int_C dz$, where C is the left half of the ellipse $\frac{1}{36}x^2 + \frac{1}{4}y^2 = 1$ from $z = 2i$ to $z = -2i$

15. $\oint_C ze^z\,dz$, where C is the square with vertices $z = 0$, $z = 1$, $z = 1+i$, and $z = i$

16. $\int_C f(z)\,dz$, where $f(z) = \begin{cases} 2, & x < 0 \\ 6x, & x > 0 \end{cases}$ and C is the parabola $y = x^2$ from $z = -1+i$ to $z = 1+i$

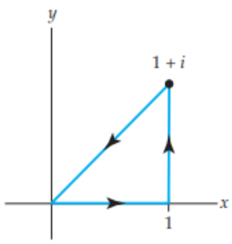

Figure 5.21 Figure for Problems 17–20

In Problems 17–20, evaluate the given integral along the contour C given in Figure 5.21.

17. $\oint_C x\,dz$

18. $\oint_C (2z-1)\,dz$

19. $\oint_C z^2\,dz$

20. $\oint_C \bar{z}^2\,dz$

In Problems 21–24, evaluate $\int_C (z^2 - z + 2)\,dz$ from i to 1 along the contour C given in the figures.

21.

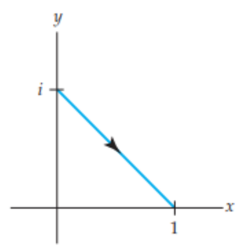

Figure 5.22 Figure for Problem 21

22.

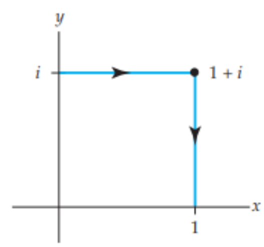

Figure 5.23 Figure for Problem 22

23.

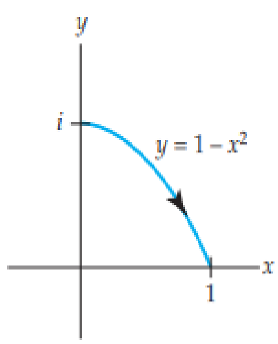

Figure 5.24 Figure for Problem 23

24.

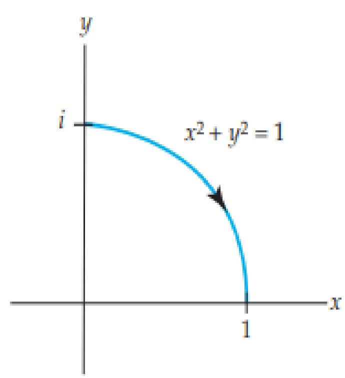

Figure 5.25 Figure for Problem 24

Additional problem to solve

1. $\int_C (e^{2z} + \cos z)\, dz$, C is the shortest path from $z = 2$ to $z = 4$.
2. $\int_C (z \cdot \bar{z})\, dz$, C is the path around the square with vertices $0, 1, 1+i, i$.
3. $\int_C \left(\frac{5}{z-2i} - \frac{6}{(z-2i)^2}\right) dz$, C is the circle $|z - 2i| = 4$, clockwise.
4. Evaluate $\int_C z^2\, dz$ where C is the straight line from $z = 0$ to $z = 3i$.
5. Evaluate $\int_C \bar{z}\, dz$ where C is the straight line from from $z = 0$ to $z = 3$.
6. $\int_C \ln(z) dz$, C is the shortest path from i to $2i$.
7. Evaluate $\int_C (z + z^{-1})dz$ along the path C, which is the circle $|z| = 2$.

Laurent Series

1. Expand $f(z) = \frac{1}{z(z-1)}$ in a Laurent series valid for
   a) $0 < |z| < 1$ b) $1 < |z|$ c) $0 < |z-1| < 1$ d) $1 < |z-1|$
2. Expand $f(z) = \frac{1}{(z-1)^2(z-3)}$ in a Laurent series valid for
   a) $0 < |z-1| < 2$ b) $0 < |z-3| < 2$
3. Expand $f(z) = \frac{8z+1}{z(1-z)}$ in a Laurent series valid for $0 < |z| < 1$
4. Expand $f(z) = \frac{1}{z(z-1)}$ in a Laurent series valid for $1 < |z-2| < 2$
5. Expand $f(z) = \frac{1}{z(z-3)}$ in a Laurent series valid for
   a) $0 < |z| < 3$ b) $|z| > 3$ c) $0 < |z-3| < 3$
   d) $|z-3| > 3$ e) $1 < |z-4| < 4$ f) $1 < |z+1| < 4$
6. Expand $f(z) = \frac{1}{(z-1)(z-2)}$ in a Laurent series valid for
   a) $1 < |z| < 2$ b) $|z| > 2$ c) $0 < |z-1| < 1$ d) $0 < |z-2| < 1$
7. Expand $f(z) = \frac{z}{(z+1)(z-2)}$ in a Laurent series valid for
   a) $0 < |z+1| < 3$ b) $|z+1| > 2$ c) $1 < |z| < 2$ d) $0 < |z-2| < 3$

8. Expand $f(z) = \frac{7z-3}{z(z-1)}$ in a Laurent series valid for

a) $0 < |z| < 1$ b) $0 < |z-1| < 1$

9. Expand $f(z) = \frac{z^2-2z+2}{z-2}$ in a Laurent series valid for

a) $1 < |z-1|$ b) $0 < |z-2|$

Additional problem to solve

1. Expand $f(z) = \dfrac{1}{(1+z^2)(z+2)}$ when (i) $1 < |z| < 2$, (ii) $|z| > 2$.

2. Expand $f(z) = \dfrac{3z}{(z-1)(2-z)}$ in a Laurent series valid for

   (a) $|z| < 1$, (b) $1 < |z| < 2$, (c) $|z| > 2$, (d) $|z-1| > 2$ and (e) $0 < |z-1| < 1$

3. Expand $f(z) = \dfrac{1}{z(z-2)}$ in a Laurent series valid for

   (a) $0 < |z| < 2$ and (b) $|z| > 2$

4. Expand $f(z) = \dfrac{5z}{(z^2+1)(z+2)}$ in a Laurent series valid for

   (a) $1 < |z| < 2$ and (b) $|z| > 2$

Zeros and Poles

1. Determine the poles and order of the poles for the given function

a) $f(z) = \frac{2z+5}{(z-1)(z+5)(z-2)^4}$

b) $f(z) = \frac{3z-1}{z^2+2z+5}$

c) $f(z) = 5 - \frac{6}{z^2}$

d) $f(z) = \frac{1+4i}{(z+2)(z+i)^4}$

e) $f(z) = \frac{z-1}{(z+1)(z^3+1)}$

f) $f(z) = \frac{\cot \pi z}{z^2}$

g) $f(z) = \frac{1-\cosh z}{z^4}$

h) $f(z) = \frac{e^z}{z^2}$

i) $f(z) = \frac{e^z-1}{z^2}$

j) $f(z) = \frac{\sin z}{z^2-z}$

k) $f(z) = \frac{\cos z - \cos 2z}{z^6}$

Residues and Residue Theorem

1. Find the singular point and its corresponding residues of the following functions

a) $f(z) = \frac{1}{(z-1)^2(z-3)}$ d) $f(z) = \frac{4z+8}{2z-1}$ g) $f(z) = \frac{5z^2-4z+3}{(z+1)(z+2)(z+3)}$

b) $f(z) = \frac{1}{z^4+1}$ e) $f(z) = \frac{1}{z^4+z^3-2z^2}$ h) $f(z) = \frac{2z-1}{(z-1)^4(z+3)}$

c) $f(z) = \frac{1}{z^2+16}$ f) $f(z) = \frac{1}{(z^2-2z+2)^2}$ i) $f(z) = \frac{\cos z}{z^2(z-\pi)^2}$

2. Evaluate the followings applying Cauchy's residue theorem (CRT)

a) $\oint_C \frac{1}{(z-1)^2(z-3)} dz$; where

i) the contour C is the rectangle defined by $x = 0, x = 4, y = -1, y = 1$

ii) and the contour C is the circle $|z| = 2$

b) $\oint_C \frac{2z+6}{z^2+4} dz$; where the contour C is the circle $|z - i| = 2$

c) $\oint_C \frac{e^z}{z^4+5z^3} dz$; where the contour C is the circle $|z| = 2$

d) $\oint_C \frac{1}{(z-1)(z+2)^2} dz$; where where the contour C is i) $|z| = \frac{1}{2}$ ii) $|z| = \frac{3}{2}$ ii) $|z| = 3$

e) $\oint_C \frac{z+1}{z^2(z-2i)} dz$; where where the contour C is i) $|z| = 1$ ii) $|z - 2i| = 1$ ii) $|z - 2i| = 3$

f) $\oint_C \frac{1}{z^2+4z+13} dz$; where the contour C is $|z - 3i| = 3$

g) $\oint_C \frac{1}{z^3(z-1)^4} dz$; where the contour C is $|z - 2| = \frac{3}{2}$

h) $\oint_C \frac{z}{z^4-1} dz$; where the contour C is $|z| = 2$

i) $\oint_C \frac{z}{(z+1)(z^2+1)} dz$; where the contour C is $16x^2 + y^2 = 4$

j) $\oint_C \frac{ze^z}{z^2-1} dz$; where the contour C is $|z| = 2$

k) $\oint_C \frac{e^z}{z^3+2z^2} dz$; where the contour C is $|z| = 3$

l) $\oint_C \frac{\tan z}{z} dz$; where the contour C is $|z - 1| = 2$

m) $\oint_C \frac{\cot \pi z}{z^2} dz$; where the contour C is $|z| = \frac{1}{2}$

n) $\oint_C \frac{\cos z}{(z-1)^2(z^2+9)} dz$; where the contour C is $|z - 1| = 1$

o) $\oint_C \frac{1}{z^6+1} dz$; where the contour C is the semicircle defined by $y = 0, y = \sqrt{4 - x^2}$

Additional problem to solve

a) $\oint_C \frac{dz}{z-3i}$, where C is the circle $|z| = 4$.

b) $\oint_C \frac{e^{-z}}{(z-1)^2}$, where C consists of $|z| = 4$.

c) $\oint_C \frac{dz}{(z-6)^{10}}$, where C is the circle $|z| = 4$.

d)) $\oint_C \frac{dz}{z^2+4}$; , C is the contour as (i) $|z+2\,i|=1$

e) $\oint_C \frac{cos(\pi z^3)}{(z-1)(z-2)} dz$; where C is the circle $|z-3|=4$,

f) $\oint_C \frac{sin\,3z}{(z-\pi)^2} dz$; where C is the circle $|z|=4$.

g) $\oint_C \frac{z+2}{z-2} dz$; where C is the circle $|z-1|=2$

h) $\oint_C \frac{\sin \pi z}{(z-2)^2} dz$; where C is the circle $|z|=3$.

i) $\oint_C \frac{dz}{z^3}$; where C is the circle $|z+1|=3$.

Evaluation of Real Improper Integrals

1. Evaluate the following improper integral using Cauchy's residue theorem (CRT):

a) $\int_{-\infty}^{\infty} \frac{1}{(x^2+1)(x^2+9)}\,dx$ b) $\int_{-\infty}^{\infty} \frac{1}{x^2+1}\,dx$ c) $\int_{0}^{\infty} \frac{x\,sin\,x}{x^2+9}\,dx$

d) $\int_{-\infty}^{\infty} \frac{1}{x^2-2x+2}\,dx$ e) $\int_{-\infty}^{\infty} \frac{1}{x^2-6x+25}\,dx$ f) $\int_{-\infty}^{\infty} \frac{1}{(x^2+4)^2}\,dx$

g) $\int_{-\infty}^{\infty} \frac{x^2}{(x^2+1)^2}\,dx$ h) $\int_{-\infty}^{\infty} \frac{1}{(x^2+1)^3}\,dx$ i) $\int_{-\infty}^{\infty} \frac{x}{(x^2+4)^3}\,dx$

j) $\int_{-\infty}^{\infty} \frac{2x^2-1}{x^4+5x^2+4}\,dx$ k) $\int_{-\infty}^{\infty} \frac{1}{(x^2+1)^2(x^2+9)}\,dx$ l) $\int_{0}^{\infty} \frac{x^2}{x^2+1}\,dx$

m) $\int_{0}^{\infty} \frac{1}{x^2+1}\,dx$ n) $\int_{-\infty}^{\infty} \frac{x^2}{(x^2-2x+2)(x^2+1)^2}\,dx$

Additional problem to solve

a) $\int_0^{\infty} \frac{dx}{\left(x^4+16\right)}$ b) $\int_{-\infty}^{\infty} \frac{dx}{x^2+2x+2}$, c) $\int_{-\infty}^{\infty} \frac{x^2}{(x^2+1)^2} dx$,

d) $\int_{-\infty}^{\infty} \frac{dx}{\left(x^2-2x+2\right)^2}$, e) $\int_0^{\infty} \frac{x^2}{x^6+1} dx$.

Z-Transformation

1. Practice

   a. $y_k = 2^k$
   b. $y_k = \left(-\frac{1}{2}\right)^k$
   c. $y_k = \delta_k(2)$
   d. $y_k = \delta_k(-3)$
   e. $y_k = u_k(3)$
   f. $y_k = u_k(5)$
   g. $y_k = k^2$

2. Find the z-transformation of each of the following-

   a. $y_k = 2 + 3k.$
   b. $y_k = k^3.$
   c. $u_k = 3y_{k+3}.$
   d. $v_k = \mathrm{k} \cdot \cos\left(\frac{k\pi}{2}\right).$

3. Find the sequences whose z-transforms are-

   a. $Y(z) = \frac{2z^2-3z}{z^2-3z-4}.$
   b. $U(z) = \frac{3z^2-4z}{z^2-3z+2}.$
   c. $V(z) = \frac{2z^2+z}{(z-1)^2}.$
   d. $V(z) = \frac{z^2+3z}{(z-3)^2}.$
   e. $W(z) = \frac{3z^2+5}{z^4}.$

4. Solve the following first order initial value problems using z-transforms.

   a. $y_{k+1} - 3y_k = 4^k,\ y_0 = 0.$
   b. $y_{k+1} + 4y_k = 10,\ y_0 = 3.$
   c. $y_{k+1} - 5y_k = 5^{k+1},\ y_0 = 0.$
   d. $y_{k+1} - 2y_k = 3 \cdot 2^k,\ y_0 = 3.$
   e. $y_{k+1} + 3y_k = 4\delta_k(2),\ y_0 = 2.$

5. Solve the following second order initial value problems using z-transforms:

   a. $y_{k+2} - 5y_{k+1} + 6y_k = 0,\ y_0 = 1,\ y_1 = 0.$
   b. $y_{k+2} - y_{k+1} - 6y_k = 0,\ y_0 = 5,\ y_1 = -5.$
   c. $y_{k+2} - 8y_{k+1} + 16y_k = 0,\ y_0 = 0,\ y_1 = 4.$
   d. $y_{k+2} - y_k = 16 \cdot 3^k,\ y_0 = 2,\ y_1 = 6.$
   e. $y_{k+2} - 3y_{k+1} + 2y_k = u_k(4),\ y_0 = 0,\ y_1 = 0.$

6. Solve the following systems using z-transforms:

a.

$$u_{k+1} - 2v_k = 2.4^k$$

$$-4u_k + v_{k+1} = 4^{k+1}$$

$$u_0 = 2, v_0 = 3$$

b.

$$u_{k+1} - v_k = 0$$

$$u_k + v_{k+1} = 0$$

$$u_0 = 0, v_0 = 1$$

c.

$$u_{k+1} - v_k = 2k$$

$$-u_k + v_{k+1} = 2k + 2$$

$$u_0 = 0, \ v_0 = 1$$

d.

$$u_{k+1} - v_k = -1$$

$$-u_k + v_{k+1} = 3$$

$$u_0 = 0, \ v_0 = 2.$$

The Z-Transformation (Additional Problem)

1) The number of comparisons needed to sort an array of n elements by the method bubble sort (or by straight selection) can be expressed by the following recurrence relation $a_{n+1} = a_n + n, n \geq 1$ with initial condition $a_1 = 0$. Solve it and check your answer by direct substitution.

2) The number of moves of disks, necessary to solve the Tower of Hanoi puzzle for n disks can be expressed by the following recurrence relation $a_{n+1} = 2a_n + 1, n \geq 1$ with initial condition $a_1 = 1$. Solve it and check your answer by direct substitution.

3) The maximum number of regions defined by n straight lines in the plane can be expressed by the following recurrence relation $a_{n+1} = a_n + n + 1,\ n \geq 0$with initial condition $a_0 = 1$. Solve it and check your answer by direct substitution.

4) The number of all squares in a square grid of dimension n can be expressed by the following recurrence relation $a_{n+1} = a_n + (n+1)^2, n \geq 1$ with initial condition $a_1 = 1$. Solve it and check your answer by direct substitution.